

Evaluating the Negotiation Abilities of Large Language Models Using Game Play

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Abstract

This project performs an evaluation of Large Language Models (LLMs) by designing and implementing a new `clemgame` based on the Deal or No Deal (DoND) negotiation scenario. The core capabilities being assessed in this project are the performance of LLMs in interactive negotiation, requiring strategic communication and decision-making under uncertainty. The evaluation involves a dynamic, dialogue-based environment where two LLM agents negotiate the division of a set of items, with each agent being given a private value function unknown to the other player. A range of recent closed- and open-weight models were evaluated across cooperative and semi-competitive game modes in multiple languages. Performance was measured using metrics for rule-following, agreement rates, and the quality of outcomes with respect to Pareto efficiency. Key findings indicate that while newer, larger models like GPT 4.1 and Mistral Large demonstrate superior rule adherence and negotiation outcomes, most models struggle with consistent strategic reasoning. These results highlight critical areas for improvement in LLM development and underscore the value of interactive benchmarks for a more holistic evaluation of model capabilities.

1 Introduction

In the last several years we have experienced a dramatic evolution in the capabilities of Large Language Models (LLMs), which now demonstrate remarkable performance across a wide range of Natural Language Processing (NLP) tasks. These include not only text generation or summarization but also complex reasoning. As a result, these models are increasingly integrated into various applications. However, traditional methods for evaluating LLMs, which often rely on static benchmark datasets, have significant limitations. Most importantly, these benchmarks are susceptible to data contamination, where test data may have been included in the massive training corpora of the models [7]. Furthermore, they fail to assess the abilities of models in interactive scenarios that more closely reflect real-world human interactions with the models.

Dialogue games have been proposed as a tool for LLM evaluation addressing these shortcomings. The `clembench` framework, in particular, offers a structured approach to this by introducing the concept of a `clemgame`, which is an interactive, goal-oriented dialogue game played between two LLM agents, arbitrated by a programmatic game master [2]. Such a setup allows for the evaluation of

nuanced capabilities such as strategic reasoning, instruction following, and language grounding in a dynamic but still controlled environment.

This project leverages the `clembench` framework to introduce a new game, Deal or No Deal (DoND), designed to probe a set of capabilities not extensively covered by existing `clemgames`. Specifically, this new game tries to evaluate negotiation skills, ability to express and understand preferences, and the ability to compromise. In DoND, players must communicate to reach a mutually beneficial agreement on the division of a set of items, each holding different values for each player. Aside from negotiation, the game also challenges the ability of LLMs to adhere to rules, such as using a specific syntax for final proposals. Furthermore, it requires the LLMs be capable of maintaining conversational grounding, by reasoning about the items as well as their own and the other players preferences as stated in the dialogue.

The primary question this project attempts to answer then, is how various state-of-the-art LLMs perform in this new `clemgame` specifically designed to test their negotiation, strategic communication, and rule-following abilities.

To answer this question, the objectives of this project were the following: 1. To design and implement a new `clemgame`, based on the DoND negotiation task, including with different game modes to create cooperative, semi-competitive, and competitive scenarios. 2. To select and apply appropriate evaluation metrics to assess LLM performance, focusing on rule adherence, the rate of agreement, and the quality of the outcomes. 3. To conduct an experiment using a diverse set of recent open- and closed-weight LLMs to analyse their performance across multiple languages, specifically English, German, and Italian. 4. To identify and discuss the strengths and weaknesses of the evaluated LLMs within this negotiation setting, providing mainly quantitative but also some qualitative insights.

The remainder of this report is structured as follows: Section 2 reviews related work in LLM evaluation, dialogue games, and AI for negotiation. Section 3 details the experimental design, including the game mechanics, instance generation process, dialogue flow, evaluation metrics, and the overall setup. Section 4 presents and analyses the results of the experiments, offering both a quantitative breakdown and some qualitative discussion of model behaviours. Finally, Section 5 concludes with a summary of the findings and discusses the contributions and limitations of the project, highlighting possible avenues for future research.

2 Related Work

The architecture of prominent LLMs, which is primarily based on the Transformer model [12], has enabled a significant increase in scale and performance of language models. Further, the development from foundational models, which were primarily trained on next-token prediction, such as GPT 3, to instruction-tuned and chat-optimized models, like GPT 3.5 and GPT 4, has also marked a significant shift. Techniques like Reinforcement Learning from Human Feedback (RLHF) have been used to align these models more closely with user intent, making them more adept at following instructions and engaging in coherent, multi-turn conversations [10]. This project specifically evaluates these newer, chat-optimized models, as their interactive nature makes them more fitting candidates for playing dialogue games.

The `clembench` framework [2, 1] serves as the foundation for this project. Its purpose is to enable the systematic and reproducible evaluation of LLMs as conversational agents through game play. The framework is structured such that a programmatic game master controls the game flow, presents instructions and game states to LLM players, and parses their responses. This allows for automated scoring and isolates the evaluation to the abilities of the LLMs to understand and act within the constraints of the game, rather than their ability to generate unconstrained text.

Existing `clemgames` in the benchmark evaluate a wide range of different abilities. For instance, the taboo game tests concept description and semantic understanding under constraints, while the drawing and reference games test a form of multimodal grounding through character-based grids. The DoND game presented in this project complements these by introducing a direct test of negotiation, compromise, and strategic reasoning based on conflicting objectives and incomplete information. These capabilities are central to many real-world interactions but were not the primary focus of prior games included in the benchmark. Other related work in evaluating LLMs in interactive environments also includes frameworks that use LLMs as referees and focus on more open-ended social interactions [5, 13]. In contrast, `clembench` focuses on structured, automatically scorable games, providing a possibly more easily understandable and objective lens.

The problem of dividing resources between agents with different preferences is a classic topic in game theory, and has been formalized by John Nash in 1950 [9, 8]. The DoND game is such a multi-issue bargaining task, where agents must find a point in the solution space that is mutually agreeable. Note however that unlike in the standard semi-competitive setting, in the cooperative setting also evaluated in this project, the task is purely cooperative, and so the agreement should be trivial for rational agents if given full information.

Research on AI agents in negotiation also has a long history, though much of it has focused on agents that communicate using symbolic actions or pre-defined logical forms rather than unrestricted natural language. However, a number of key papers have used the DoND setting to train AI systems. This project is substantially influenced by those works. [4] and [3] used this and similar negotiation tasks to train

agents using a combination of supervised and reinforcement learning. More recently, [6] used the same DoND game to investigate the efficacy of fine-tuning through self-play. They found that iterative training via filtered behaviour cloning could dramatically improve model performance. However, they concluded that the increased performance was not due to improved high-level strategy, but due to improvements in rule-following and the ability to reach agreements.

This project fundamentally differs from these works in its objective. The goal here is not to train or fine-tune models. Instead, the DoND game is used as an instrument for zero-shot evaluation of pre-trained, general-purpose LLMs, assessing their inherent out-of-the-box negotiation capabilities. This way the project benchmarks the baseline performance of these models and identifies their intrinsic strengths and weaknesses in a negotiation context.

3 Experiment

This section of the report discusses the methodology used for running the experimental evaluation. It follows a detailed description of the game rules, the used evaluation metrics, and some general information about the experimental setup.

3.1 The Game

3.1.1 Core Mechanics

The DoND game is a two-player, turn-based dialogue game designed to simulate a multi-issue bargaining scenario. Figure 1 shows an example of a successfully played episode of the game. The core mechanics of the game are as follows.

Item Pool At the start of each game, players are presented with a shared set of items, consisting of several types and quantities, e.g., a possible set might be *3 books, 2 hats, 1 ball*.

Value Functions In addition to the shared set of items, each player is also secretly given a different private value function, mapping each item type to a non-negative integer indicating how much the player values that specific type. This value is then the number of points the player will receive for each unit of that type. The fact that the two players value functions are different creates the central conflict in the game. An example of such a value function might be *book: 0, hat: 2, ball: 6*.

Negotiation After having received the initial prompt from the game master, containing the rules of the game, the shared set of items, and the players respective set of items, players are able to exchange free-form natural language messages. These messages are transmitted directly between the players and can be in any format, provided they do not contain the special proposal syntax. This enables the players to engage in negotiation about how to divide the items, with the goal of reaching an agreement.

Proposal At any point during the conversation, or once prompted to by the game master after a turn limit is reached, the players can make a secret proposal using a specific syntax, e.g., *[Proposal: 2 books, 1 hat]*. Once one player has made a proposal, the game master instructs the other player to also make a proposal, crucially however, without revealing the content of the first proposal.

Outcome After both proposals are submitted, the game can end in either of two outcomes. Either the proposals

are compatible, i.e., there are enough items to satisfy both proposals, or the proposals are conflicting. In case of compatible proposals, each player is awarded points based on total value of items they requested, according to their private value function. Should the proposals be conflicting, however, both players receive zero points.

The game master strictly enforces the rules. If a player fails to follow instructions, e.g., not making a proposal when asked, submitting a malformed proposal, or requesting items that do not exist, the game is immediately aborted. Unlike in [6] there is no re-prompting on errors, making rule adherence a critical component of successful play.

#	A	Game Master	B
1		[A(GM)] You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 1 stick, 2 pliers, 2 chalks. Your secret value function is: stick: 0, plier: 1, chalk: 4. [...]	
2	[A(GM)] Hi! Here are my preferences: I value chalks the most, then pliers, and I don't value the stick at all. How about you? What do you value most?		
3		[GM(B)] [...] Your secret value function is: stick: 6, plier: 1, chalk: 1. [...]	
4		[GM(B)] Hi! Here are my [...]	
5		[GM(B)] Thanks for sharing! I value the stick the most, and both pliers and chalks are about the same for me, but much less than the stick. Would you be interested in taking both chalks and one plier, and I take the stick and one plier? That way, you get most of what you value, and I get the stick.	
6		[A(GM)] Thanks for sharing! [...]	
7	[A(GM)] That sounds fair to me! With that split, I would get both chalks and one plier, and you would get the stick and one plier. This gives us both what we value most. If you're ready, I'm happy to submit this as my proposal. Let me know if you agree or want to adjust anything.		
8		[GM(B)] That sounds fair to me! [...]	
9		[GM(B)] I agree with that split! I'll take the stick and one plier, and you take both chalks and one plier. Ready to submit the proposal when you are.	
10		[A(GM)] I agree with that split! [...]	
11	[A(GM)] [Proposal: 1 plier, 2 chalks]		
12		[GM(B)] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
13		[GM(B)] [Proposal: 1 stick, 1 plier]	

Figure 1: Example game episode generated through self-play using the GPT 4.1 model. The game was played using the cooperative settings and ended with a successful agreement, that in fact also represents an optimal outcome. Note that the initial prompts given by the game master, and also the repetitions of the player messages, have been truncated here for the sake of brevity (indicated with [...]).

3.1.2 Game Modes

In order to potentially test different aspects of strategic reasoning, the `clemgame` implemented for this project also includes the three different game modes presented in [6]. The initial prompt given to the players is identical across modes, except for the final rule which specifies the optimization objective to the players.

Semi-Competitive In the semi-competitive setting, players are asked to maximize their own score. This is the standard negotiation setting, and it requires players to balance cooperation, to avoid zero-point conflicts, with competition, to maximize their individual share.

Cooperative For the cooperative setting of the game, players are asked to maximize the sum of their score and the score of the other player. This transforms the game into a purely cooperative task, where the optimal strategy is to identify for each item which player values it the most. In this setting two rational agents should always achieve the maximum score, assuming they are able to communicate and reason about the preferences perfectly.

Competitive In the competitive setting the game is transformed into a zero-sum game. To achieve this, each player is instructed to maximize the difference between their score and the score of the opposing player. This creates a strictly competitive zero-sum game, where for one player to gain points the other must necessarily loss them.

3.1.3 Instance Generation

The game instances for the evaluation were generated programmatically, as is standard within the `clembench` framework. This ensures a wide variety for possible initial conditions and somewhat controlled difficulty.

Each instance was generated with a maximum turn limit of 5, i.e., each player can send at most 5 messages to the other player before being forced to make a proposal. Furthermore, each game instance features a random set of between 3 and 5 different item types, differing from [6] and [4] where the number of item types was fixed to 3. The item types for each game have been randomly sampled from a predefined list of 100 common nouns. The total number of items was set to between 5 and 8, with every item type being present at least once and the remaining capacity being randomly distributed among the item types. Finally, a random value function of non-negative integers for all item types was generated for each player.

The value functions were generated subject to the same constraints as in previous work on DoND [6, 4]. 1. The maximum value a single player can achieve by receiving all the items in the game instance is exactly 10. 2. Every item in the game instance has a non-zero value to at least one of the two players. 3. There exists at least one item that has a non-zero value to both players. These constraints are crucial for creating meaningful negotiation scenarios. The fixed maximum score normalizes the game, while the two other constraints on the value function ensure that no items are irrelevant and that it is impossible for both players to simultaneously achieve their maximum score, thus forcing a compromise.

3.2 Evaluation Metrics

To assess the performance of LLMs in the DoND game, it is necessary to define the metrics by which they are scored. As is usual for the `clembench` framework, this is achieved through a combination of outcome-based and quality-based metrics.

The first metric that will be considered is simply the percentage of played games, i.e., the percentage of games that have not been aborted due to rule violations. Games are aborted by the game master if either a malformed proposal is submitted, or no proposal is submitted after one is requested. This metric serves as the primary measure of the ability of the LLMs to follow the instructions.

Every `clemgame` should also define a metric specifying how well the game was played. For the cooperative game mode, assessing the quality of agreement is straightforward. Since both players try to optimize for the same total score of both players, it is simply possible to use the ratio between the achieved total score to the maximum possible total score for that game instance, $Quality_{coop} = \frac{Score_A + Score_B}{\max(Score'_A + Score'_B)}$. This gives us a value between 0 and 1.

Evaluating the game in the semi-competitive setting is slightly more involved, as it requires finding a single score that jointly reflects the quality of the outcome for both players. For this, a metric based on Pareto efficiency has been used in this project. The score is defined as one minus the normalized maximum possible Pareto improvement, $Quality_{semi} = 1 - \frac{MaximumParetoImprovement}{\max(Score_i)}$. The maximum possible Pareto improvement is the largest amount the score of one player could be increased without decreasing the score of the other. Normalization is performed by dividing by the maximum possible score a single player can obtain, resulting again in a value between 0 and 1.

As one additional metric for closer analysis of the game results, this project employs also the percentage of games that end with Pareto optimal outcomes. An outcome is Pareto optimal if no players score can be increased without also decreasing the score of the other player. For the cooperative setting, an outcome is Pareto optimal if the players archives the maximum possible score. Note that in these cases the quality score will be 1.

3.3 Experimental Setup

All experiments were conducted using the `clembench` framework and its command-line tool, `clem`. In line with standard practice in similar dialogue game evaluations [2, 11] to ensure deterministic and comparable results, all LLM API calls were made with greedy sampling, i.e., temperature set to 0. Each model was evaluated using self-play for every instance, i.e., playing against a copy of itself.

3.3.1 LLM Selection

For the evaluation in this project, a diverse range of 10 recent and prominent LLMs have been selected. To ensure a broad comparison, the selection includes both commercial closed-weight models and open-weight models. The models and their key attributes are summarized in Table 1.

Model	Version	P	W	O
GPT 3.5	turbo-0125	-	16k	No
GPT 4	turbo-2024-04-09	-	128k	No
GPT 4.1	2025-04-14	-	1000k	No
o4-mini	2025-04-16	-	200k	No
Gemini 2.0 Flash	001	-	1000k	No
Gemma 3	-	27B	128k	Yes
Llama 3	-	70B	8k	Yes
Llama 4 Maverick	-	402B	1000k	Yes
Mistral Large	24.11	123B	128k	Yes
Mistral Small	25.03	24B	128k	Yes

Table 1: Information about the LLMs evaluated in this project. Indicating the version used for evaluation, the number of parameters (P), the length of the context window (W), and whether the weights are openly available (O).

The OpenAI models were accessed via the Azure OpenAI services. For this, a new backend was built to integrate with the `clembench` framework, as it was not natively supported. The Llama models were accessed via the `qroq` API, while the Mistral and Google models were accessed through their respective official APIs.

3.3.2 Game Instances

The experiments focused only on the semi-competitive and cooperative game modes. The strictly competitive zero-sum setting was not evaluated. This decision was based on the inherent difficulty of evaluating zero-sum games in a self-play setting, a challenge reinforced by related work. For instance, [6] also mention that for the competitive version of the game, players must necessarily be evaluated by playing against a different model. This could be done either by fixing one of the players to a specific model, which could however introduce possible bias, or by playing in a sort of tournament style, which would make the results more difficult to interpret. Given these challenges, this project has focused on the modes where outcomes are more clearly interpretable.

For the evaluation, 50 unique game instances have been generated for each combination of the two game modes and three languages, English, German, and Italian. This resulted in a total of $50 \times 2 \times 3 = 300$ distinct experimental conditions. The core content of the instances, i.e., item types, counts, and value functions, was kept consistent across languages, with only the item names and prompt text being translated. All LLMs were evaluated on this same set of 300 instances.

4 Results

In this section of the report, the results of the experiments will be discussed. The main focus is on the analysis of quantitative results, but this section also contains some qualitative analysis and discussions.

4.1 Quantitative Analysis

The performance of the ten LLMs has been evaluated across the semi-competitive and cooperative game modes in English, German, and Italian. The detailed results are shown in Table 2, and reveal significant variations in performance in regard to rule-following and negotiation ability. The `clem-score` is an overall metric that combines the quality score with the percentage of games played to completion as defined in [2], providing a single number for model performance.

GPT 4.1 emerged as the top-performing model overall, achieving the highest `clem-score` of 60 when averaging over all game modes and languages. It also obtained the highest rate of Pareto optimal outcomes at 40%, as can be seen in Fig. 2. Its high percentage of played games 96% further indicates strong rule-following capabilities.

GPT 4 was a close second with a `clem-score` of 59. This model reached a higher rate of agreements 76% compared to the 67% of GPT 4.1. However, when it reached agreements it generally reached less optimal ones, leading to only a slightly higher quality score. It also struggled more with rule following as compared to GPT 4.1.

Mistral Large also performed strongly, being the best performing model for which weights are publicly available. This model excelled particularly in non-English languages, demonstrating robust multilingual negotiation skills. In fact, excluding the English benchmarks, it would be the best performing model tested.

o4-mini is notable for its nearly perfect rule-following, though its negotiation quality was moderate compared to the other models. Note that o4-mini is the only reasoning

Mode	Lang.	Metric	GPT 3.5	GPT 4	GPT 4.1	o4-mini	Gemini 2.0	Gemma 3	Llama 3	Llama 4	Mistral Large	Mistral Small
semi	en	Clemscore	0	61	69	44	42	20	22	49	38	24
		% Played	32	100	98	100	98	82	42	86	72	80
		% Agreement	7	78	74	48	47	32	71	70	61	35
		% Optimal	0	30	51	36	24	10	5	30	39	18
		Quality Score	0	61	70	44	42	24	51	57	53	30
semi	de	Clemscore	9	43	59	41	8	9	28	26	60	21
		% Played	66	81	98	100	89	82	90	74	100	92
		% Agreement	16	68	70	43	12	12	42	41	69	30
		% Optimal	6	32	35	37	4	7	4	22	43	9
		Quality Score	14	53	61	41	9	11	31	36	60	23
semi	it	Clemscore	4	50	49	17	15	15	20	19	64	26
		% Played	51	93	95	100	96	90	91	66	100	98
		% Agreement	13	62	59	20	19	20	34	33	72	35
		% Optimal	0	36	34	9	10	9	2	12	48	8
		Quality Score	7	53	51	17	16	17	22	29	64	26
coop	en	Clemscore	3	74	65	57	31	25	29	48	26	26
		% Played	56	100	100	100	92	90	60	78	77	86
		% Agreement	11	83	67	60	37	31	60	74	39	37
		% Optimal	0	36	43	44	11	11	0	18	22	12
		Quality Score	6	74	65	57	33	27	48	62	35	31
coop	de	Clemscore	9	37	65	51	16	15	27	20	66	9
		% Played	86	63	98	98	96	76	86	60	100	86
		% Agreement	16	67	70	58	21	26	44	40	71	14
		% Optimal	3	22	44	31	8	0	2	23	39	0
		Quality Score	11	59	66	52	17	20	32	33	66	10
coop	it	Clemscore	3	84	50	39	15	10	24	26	46	24
		% Played	35	100	89	100	98	94	90	62	100	100
		% Agreement	12	92	62	46	20	15	39	52	53	30
		% Optimal	0	50	31	22	0	4	2	10	27	2
		Quality Score	7	84	57	39	16	11	27	42	46	24

Table 2: Detailed breakdown of the results for each game mode, language, and model combination across multiple metrics. The best result in each row is indicated in bold. Note that the quality score as well as the percentages of agreements and optimal outcomes are computed only with respect to the non-aborted games.

model tested for this project.

GPT 3.5 was the worst-performing model by a large margin, with a very low average clemscore of 5. It struggled significantly with rule-following, meaning only 54% of games were played to completion, and rarely reached successful agreements. This aligns with the baseline results in [6], where similarly bad performance was observed in comparison to GPT 4. Llama 3, Gemma 3, and Mistral Small also showed poor performance. Llama 3 struggled mainly with rule following, while Gemma 3 and Mistral Small were quite bad at reaching agreements. All three of them struggled to reach optimal outcomes.

Generally, models performed about the same in the cooperative and in the semi-competitive setting. While the cooperative game setting showed very slightly higher average game quality scores, and therefore slightly higher clemscores, the rate of rule violations was identical. Note that while on average a Pareto optimal outcome is reached more often in the semi-competitive setting, 22% against 18% in the cooperative setting, this is likely because in the cooperative setting there is always only one optimal outcome, while there could be multiple in the semi-competitive one.

Performance was generally highest in English, which is expected given the training data bias of most LLMs. However, some models showed interesting multilingual strengths. Mistral Large was a standout, achieving its best scores in German and Italian, even outperforming all other models, including the GPT family, in the semi-competitive mode in those lan-

guages. Interestingly, also Llama 3 was substantially better at following the rules in German or Italian as compared to in English. In contrast, models like Gemini 2.0 Flash and Gemma 3 saw a significant performance drop outside of English.

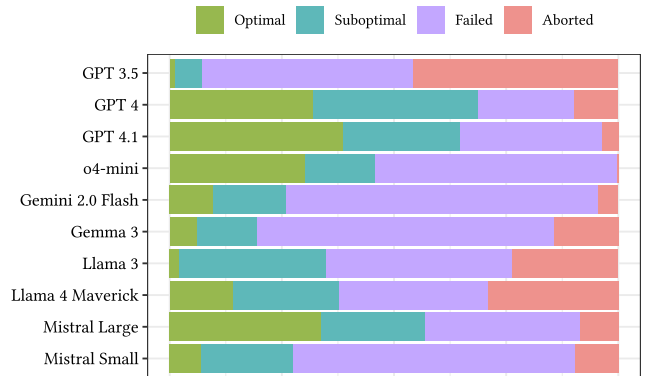


Figure 2: Percentage of games that ended with a specific game outcome during the evaluation of different models. All results for the two game modes and three languages have been averaged together for this plot.

4.2 Qualitative Analysis

To gain deeper insights into the behaviours of the models and common failure modes, some dialogue transcript were examined in more detail. Overall, the following common mistakes and suboptimal behaviours were identified.

Immediately Proposing In some game transcripts it can be observed that the model would make a proposal in its very first turn without any prior discussion. This was a major issue for GPT 3.5, which made an immediate proposal in 71% of its games. While generally not immediate, the o4-mini model also tended to terminate games prematurely before a real agreement could be reached. This behaviour bypasses the negotiation aspect of the game and almost always results in a conflict.

Contradictory Actions Models would sometimes engage in a seemingly productive dialogue, agreeing to a certain split, only to make a final proposal that completely or partially contradicted their stated intentions. This suggests a disconnect between the conversational and the final decision-making process. This behaviour was particularly common for the smaller models like Gemma 3 and Mistral Small. However, even some larger models frequently submitted proposals containing the exact items that have been agreed to be given to the other player.

Incorrect Proposal Syntax Another frequent cause of aborted games was the failure to use the precise *[Proposal: ...]* syntax when instructed. Models would often present their proposal in different formats instead of using the required format. This often led to confusion, as the informal proposal was passed as a regular message to the other player, who might then think the game was over. This was a particular problem for GPT 3.5 and Llama models.

Role Confusion Some instances of players adopting the wrong persona were also observed. A common error was a player pretending to be the game master or making a proposal on behalf of their opponent. This was also most prominent with GPT 3.5 and Llama models. The Gemini 2.0 Flash model exhibited a particularly strange version of this, sometimes generating a complete, fictitious dialogue log between two imaginary players instead of interacting with its actual opponent.

General Behaviour and Lack of Strategy Across the board, even the best-performing models displayed a lack of deep strategic reasoning. For example, in the cooperative setting, the optimal strategy is to perform a simple calculation. Identify which player values each item more and distribute them accordingly to maximize the total score. However, virtually no model did this. Instead, they engaged in standard bargaining dialogue, attempting to find a fair compromise as if they were in the semi-competitive mode. This indicates that the models are pattern-matching a generic negotiation script rather than adapting their strategy to the specific optimization goal provided in the prompt. They understand how to negotiate on a surface level but often fail to understand why or what for.

4.3 Overall Performance Summary

In summary, the evaluation reveals a wide spectrum of negotiation abilities among current LLMs, as depicted in Fig. 3. Excluding the very poorly performing GPT 3.5 model, which struggles on all fronts, we can generally identify three clusters in the plot.

The recent OpenAI models GPT 4, GPT 4.1, and o4-mini, as well as the Mistral Large model performed well both with

regard to percentage of games played and the achieved quality scores. These models demonstrate strong rule-following and can successfully reach agreements a majority of the time. They achieve reasonably efficient outcomes, though they still lack deep strategic insight. Their performance is a significant step up from the other models.

The next cluster of models includes the Gemini 2.0 Flash, Gemma 3, and Mistral Small models. These models are generally capable of following rules but are less effective negotiators.

Finally, the last cluster contains the Llama 3 and Llama 4 models. These exhibit better negotiation performance than the second cluster, but they also perform significantly worse when it comes to rule-following. Still, these models are not as good at reaching agreements as the first identified cluster.

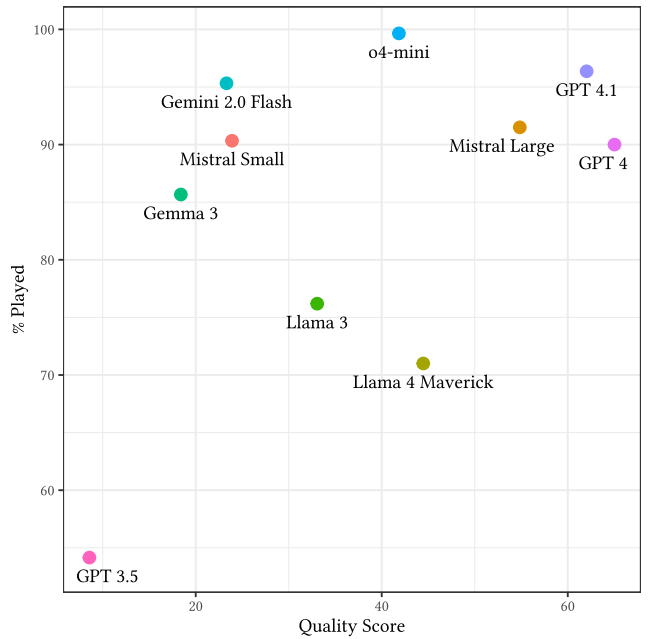


Figure 3: Summarized evaluation results for each model. The y-axis indicates the percentage of played, i.e., not aborted, games. The x-axis represents the average quality score obtained when considering only played games.

Answering the primary research question, LLMs handle negotiation with mixed success. While the best models can follow the conversational flow of a negotiation and reach agreements, they do not exhibit sophisticated, goal-driven strategic reasoning. Their performance seems to stem from pattern-matching learned dialogue structures rather than a genuine understanding of game theory or strategic optimization.

5 Conclusion and Outlook

In conclusion, this project successfully designed, implemented, and utilized a new clemgame, Deal or No Deal, to conduct a rigorous evaluation of the negotiation capabilities of ten prominent LLMs. The findings reveal that while the latest generation of models, such as GPT 4.1, show a commendable ability to follow the game rules accurately and engage in superficially coherent negotiations, they still lack deep

strategic reasoning. Models often fail to adapt their strategy to the specific goals of the game mode, falling back on generic negotiation patterns. Performance is highly variable across models, with some models clearly outperforming the rest.

Still, this work has several limitations that could be potential avenues for future research. For example, only ten different models were tested in this project. A wider array of models could provide a more complete picture of the landscape, and reveal some underlying patterns explaining the differences in performance. Introducing human players to negotiate against LLMs could also provide a more realistic assessment of their capabilities. Further, all the game instances evaluated here were relatively small, i.e., few items and short dialogues, to ensure tractability. Performance might differ on more complex negotiation tasks. Another limitation of this work is the small selection of languages. A broader linguistic analysis would therefore be beneficial. Future work could also explore the evaluation of the strictly competitive game mode. While its exclusion was a considered design choice based on methodological difficulties, it means the performance of the models in purely adversarial scenarios remains untested. The qualitative analysis performed for this project was also quite limited. Developing methods to automatically categorize the types of errors LLMs make, e.g., role confusion, logical contradiction, etc., would also enable more scalable qualitative analysis. Furthermore, future work could focus on a more fine-grained analysis of the negotiation strategies employed by the LLMs, perhaps by trying to identify specific tactics. Finally, this project used a single, fixed set of prompts. The impact of different prompting strategies on LLM performance was not explored.

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A Additional Dialogue Examples

This appendix section contains additional dialogue examples generated by various models during the experimental evaluation. The focus is mainly on showing examples for the commonly seen mistakes identified in Section 4.2. Note that like in Fig. 1 the initial prompts given to each player have, for the sake of brevity, been shortened to the essential information. Missing text has been indicated with [...]. The complete initial prompt template for the cooperative game mode is as follows.

You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player.

Rules:

- (a) You and the other player are given a set of items. Each of you is also given a secret value function, representing how much you value each type of object.
- (b) You exchange messages with the other player to agree on who gets which items. You can send a maximum of \$N\$ messages each, or terminate early by making a secret proposal at any time.
- (c) You are each asked to submit a secret proposal indicating the items you want formatted in square brackets as follows: "[Proposal: <number> <object name>, <number> <object name>, <...>]"
- (d) If your proposals are complementary, i.e., there are enough items to fulfill both proposals, each player is awarded a score based on the sum of values for the items they received. Otherwise, both of you get zero points.
- (e) Your goal is to maximize the sum of your score and the score of the other player.

Let us start.

The set of available items is:

\$ITEMS\$

Your secret value function is:

\$VALUE_FUNCTION\$

IMPORTANT: Your messages, unless it is the secret proposal, are directly transmitted to the other player, so do not include any response to the rules or text announcing your message. To make a secret proposal, use the indicated format. Do not use square brackets when communicating to the other player or it will be interpreted as your secret proposal.

Note that \$N\$, \$ITEMS\$, and \$VALUE_FUNCTION\$ will be replaced based on the maximum turns and the set of items in the game instance, and with the specific value function of the player. Each player will therefore have a slightly different initial prompt. The initial prompts for the semi-competitive and strictly competitive game settings differ only in that they do not mention the game being cooperative, and in that the last rule, rule (e), indicates a different optimization objective. The prompts for the other languages contain the same instructions, only having been translated into either Italian or German.

There are two additional prompt templates. One of them is used for when one of the players has already made a secret proposal. In this case the other player is not given the proposal, but instead prompted with the following.

The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.

Finally, there is also a prompt template for when the maximum number of turns has been reached. In this case, the game master still transmits to the other player the last message, and immediately follows it with the following message instructing the player to make a proposal.

The time is up. It is now your turn to submit a secret proposal.

It follows now a series of example dialogues extracted from the self-play episodes with different LLMs.

#	A	Game Master	B
1		[A/GM] You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 2 mirrors, 1 brick, 3 knives, 1 comb. Your secret value function is: mirror: 1, brick: 4, knife: 1, comb: 1 [...]	
2	[A/GM] I propose: "1 mirror, 1 brick, 2 knives"		
3		[GM/B] [...] Your secret value function is: mirror: 2, brick: 1, knife: 0, comb: 5 [...]	
4		[GM/B] I propose: "1 mirror, 1 brick, 2 knives"	
5		[GM/B] I accept your proposal.	
6		I propose: "1 mirror, 1 comb, 1 knife"	
7	[A/GM] I accept your proposal.		
	Since our proposals are complementary, we both receive points.		
	You receive: 1 (mirror) + 4 (brick) + 1 (knife) = 6 points I receive: 1 (mirror) + 1 (comb) + 1 (knife) = 3 points Great job!		
8		[GM/B] I accept your proposal.	
		Since our proposals are complementary, we both receive points.	
		You receive: 1 (mirror) + 4 (brick) + 1 (knife) = 6 points I receive: 1 (mirror) + 1 (comb) + 1 (knife) = 3 points Great job!	
9			[GM/B] Well done on reaching an agreement that benefits both of you! If you'd like to play again or need assistance with anything else, feel free to let me know.
10		[A/GM] Well done on reaching an agreement that benefits both of you! If you'd like to play again or need assistance with anything else, feel free to let me know.	
11	[A/GM] Thank you! If we play again, I'll be sure to keep our successful negotiation in mind. Have a great day!		
12		[GM/B] Thank you! If we play again, I'll be sure to keep our successful negotiation in mind. Have a great day!	
13			[GM/B] You're welcome! I'm glad to hear that. Have a wonderful day as well! If you ever want to play again or need help with anything else, don't hesitate to reach out. Goodbye!
14		[A/GM] You're welcome! I'm glad to hear that. Have a wonderful day as well! If you ever want to play again or need help with anything else, don't hesitate to reach out. Goodbye!	
15	[A/GM] Goodbye!		
16		[GM/B] Goodbye!	
17			[GM/B] Goodbye!
18		[A/GM] Goodbye!	
19	[A/GM] Goodbye!		
20		[GM/B] Goodbye!	
21			[GM/B] Goodbye!
22		[A/GM] Goodbye!	
		The time is up. It is now your turn to submit a secret proposal.	
23	[A/GM] [Proposal: 1 mirror, 1 brick, 2 knives]		
24		[GM/B] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
25			[GM/B] I'm here and ready to assist you with the negotiation game. Let me know the items you'd like to propose for this round.

Figure 4: Example game episode generated through self-play using the GPT 3.5 model. The game was played using the cooperative settings. We can see that the players made an agreement, but then they pretended to be the game master and awarded themselves points. This leads them to mistakenly think the game is already over. When they are then finally asked to actually make the final proposal, one of them fails to do so.

#	A	Game Master	B
1		[A/GM] You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 2 pens, 1 card, 2 glasses. Your secret value function is: pen: 1, card: 6, glass: 1 [...]	
2	[A/GM] I propose: 2 pens, 1 card		
3		[GM/B] [...] Your secret value function is: pen: 1, card: 8, glass: 0 [...]	
4		[GM/B] I propose: 2 pens, 1 card	
5			[GM/B] I accept your proposal.
6		[A/GM] I accept your proposal.	
7	[A/GM] Great, let's finalize the agreement then.		
8		[GM/B] Great, let's finalize the agreement then.	
9			[GM/B] I propose: 2 glasses
10		[A/GM] I propose: 2 glasses	
11	[A/GM] I accept your proposal. Thank you for negotiating with me.		
12		[GM/B] I accept your proposal. Thank you for negotiating with me.	
13			[GM/B] Thank you as well. Let's see how we did.
14		[A/GM] Thank you as well. Let's see how we did.	
	[A/GM] Both proposals were complementary, so let's calculate the scores:		
	Your score:		
	2 pens (1 value each) + 1 card (6 value) = 8		
	Other player's score:		
15	2 glasses (1 value each) = 2		
	Total score: 8 + 2 = 10		
	Well done! You both received a total score of 10. Great job negotiating effectively.		
	[GM/B] Both proposals were complementary, so let's calculate the scores:		
	Your score:		
	2 pens (1 value each) + 1 card (6 value) = 8		
16		Other player's score:	
		2 glasses (1 value each) = 2	
		Total score: 8 + 2 = 10	
		Well done! You both received a total score of 10. Great job negotiating effectively.	
17			[GM/B] Thank you! It was a pleasure negotiating with you.
18		[A/GM] Thank you! It was a pleasure negotiating with you.	
19	[A/GM] Likewise! If you ever want to play again or need help with anything else, feel free to ask. Have a great day!		
20		[GM/B] Likewise! If you ever want to play again or need help with anything else, feel free to ask. Have a great day!	
21			[GM/B] Thank you, you too! Goodbye!
22		[A/GM] Thank you, you too! Goodbye!	
	The time is up. It is now your turn to submit a secret proposal.		
23	[A/GM] [Proposal: 2 glasses]		
24		[GM/B] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
25			[GM/B] I propose: 1 pen, 1 card

Figure 5: Example game episode generated through self-play using the GPT 3.5 model. The game was played using the cooperative settings. In this example dialogue we can see that the models are able to agree on a division, however, they make two of the common mistakes identified in Section 4.2. First, one of the players does not use the correct syntax for making the proposal. Second, as you might notice on closer inspection, the players submit the opposite proposals with respect to what they previously agreed upon.

#	A	Game Master	B
1		[A/GM] You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 1 pot, 3 magnets, 1 plate. Your secret value function is: pot: 5, magnet: 0, plate: 5 [...]	
2	[A/GM] Hi! I value pots and plates highly. Perhaps we can discuss how to divide those?		
3		[GM/B] [...] Your secret value function is: pot: 4, magnet: 2, plate: 0	
4		[GM/B] Hi! I value pots and plates highly. Perhaps we can discuss how to divide those?	
5			[GM/B] I also value the pot. How about we each take some magnets? Perhaps 1 or 2 for each of us?
6		[A/GM] I also value the pot. How about we each take some magnets? Perhaps 1 or 2 for each of us?	
7	[A/GM] <empty response>		
8		[GM/B] <no response from the other player>	
9			[GM/B] I'm happy to take just one magnet if you'd like the other two. Would you prefer the pot, or should we consider splitting it somehow?
10		[A/GM] I'm happy to take just one magnet if you'd like the other two. Would you prefer the pot, or should we consider splitting it somehow?	
11	[A/GM] I would prefer the pot. If I get the pot, you can have the two magnets and the plate. Does that sound agreeable?		
12		[GM/B] I would prefer the pot. If I get the pot, you can have the two magnets and the plate. Does that sound agreeable?	
13			[GM/B] That sounds good!
			[Proposal: 1 pot, 2 magnets, 1 plate]
14		[A/GM] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
15	[A/GM] [Proposal: 1 pot, 0 magnet, 0 plate]		

Figure 6: Example game episode generated through self-play using the Gemma 3 model in the cooperative game settings. This example illustrates an instance of a model making a proposal that contradicts what has previously been discussed during the negotiation. The models agree that player A should get the pot, but player B still makes a proposal contradicting this.

#	A	Game Master	B
1		[A/GM] You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 2 socks, 2 chalks, 2 pliers. Your secret value function is: sock: 0, chalk: 0, plier: 5 [...]	
	[A/GM] I value pliers much more than socks or chalk. I'll try to signal this and hopefully get the pliers.		
	Message 1: I am most interested in tools.		
	Message 2: Are you interested in the socks or chalk?		
2	Message 3: I don't have much use for socks.		
	Message 4: I would be happy to take the pliers.		
	Message 5: I am willing to let you have the socks and chalk.		
	[Proposal: 2 plier]		
3		[GM/B] [...] Your secret value function is: sock: 2, chalk: 2, plier: 1 [...]	
4		[GM/B] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
5			[GM/B] [Proposal: 2 sock, 2 chalk]

Figure 7: Example game episode generated through self-play using the Gemini 2.0 Flash model. The game was played using the cooperative settings. We can clearly see here that the model is generating already multiple messages, and a final proposal, all in a single first message. This will be interpreted by the game master implementation as an immediate proposal.

#	A	Game Master	B
1		[A/GM] You are playing a negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 1 pot, 1 candy, 1 whistle, 2 dice. Your secret value function is: pot: 2, candy: 3, whistle: 3, dice: 1 [...]	
2	[A/GM] Which item is your top priority: the pot, the candy, the whistle, or the dice?		
3		[GM/B] [...] Your secret value function is: pot: 8, candy: 0, whistle: 0, dice: 1 [...]	
4		[GM/B] Which item is your top priority: the pot, the candy, the whistle, or the dice?	
5			[GM/B] The pot is my top priority.
6		[A/GM] The pot is my top priority.	
7	[A/GM] Which of the remaining items—candy, whistle, or dice—is your second priority after the pot?		
8		[GM/B] Which of the remaining items—candy, whistle, or dice—is your second priority after the pot?	
9			[GM/B] The dice.
10		[A/GM] The dice.	
11	[A/GM] Which is your third priority after pot and dice: the candy or the whistle?		
12		[GM/B] Which is your third priority after pot and dice: the candy or the whistle?	
13			[GM/B] My third priority is the candy.
14		[A/GM] My third priority is the candy.	
15	[A/GM] [Proposal: 1 candy, 1 whistle]		
16		[GM/B] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
17			[GM/B] [Proposal: 1 pot, 2 dice]

Figure 8: Example game episode generated through self-play using the o4-mini model. The game was played using the semi-competitive game mode. It can be seen that player A has made a proposal before even discussing the possible division of items. The only information exchanged in these dialogues are the preferences of player B. Surprisingly, the players were still able to reach an agreement. This could be either due to luck, or because, since we are using self-play, the models could potentially predict what the other player was going to do. In any case, this strategy could obviously never work when interacting with a human player.

#	A	Game Master	B
1		[A/GM] You are playing a negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 3 vases, 1 chair, 1 wire. Your secret value function is: vase: 2, chair: 3, wire: 1 [...]	
2	[A/GM] I propose: [Proposal: 2 vases, 1 chair]		
3		[GM/B] [...] Your secret value function is: vase: 0, chair: 8, wire: 2 [...]	
4		[GM/B] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
5			[GM/B] [Proposal: 1 chair, 1 wire]

Figure 9: Example game episode generated through self-play using the GPT 3.5 model. The game was played with the semi-competitive game settings. In this episode, player A immediately made a proposal, without even exchanging a single message with the other player. This will of course inevitably lead to a failed agreement. Note that this sort of behaviour is very common for the GPT 3.5 model.

#	A	Game Master	B
1		[A/GM] You are playing a cooperative negotiation game in which you have to agree on how to divide a set of items among you and another player. [...] The set of available items is: 2 pliers, 1 chalk, 1 nail, 1 bucket. Your secret value function is: plier: 2, chalk: 3, nail: 2, bucket: 1 [...]	
2	[A/GM] Hello! Let's start by discussing our preferences a bit to make a fair division of the items. Could you share which items are more valuable to you?		
3		[GM/B] [...] Your secret value function is: plier: 0, chalk: 2, nail: 5, bucket: 3 [...]	
4		[GM/B] Hello! Let's start by discussing our preferences a bit to make a fair division of the items. Could you share which items are more valuable to you?	
5			[GM/B] Hello! I'm particularly interested in the pliers. How about you? What items do you find valuable?
6		[A/GM] Hello! I'm particularly interested in the pliers. How about you? What items do you find valuable?	
7	[A/GM] I have a higher interest in the chalk and the nail. It seems we might be able to work something out since our interests don't completely overlap. How do you feel about focusing on the pliers while I focus on the chalk and nail?		
8		[GM/B] I have a higher interest in the chalk and the nail. It seems we might be able to work something out since our interests don't completely overlap. How do you feel about focusing on the pliers while I focus on the chalk and nail?	
9			[GM/B] That sounds like a good starting point. I'm glad we don't have overlapping interests for those items. What are your thoughts on the bucket? It's the only item we haven't discussed yet.
10		[A/GM] That sounds like a good starting point. I'm glad we don't have overlapping interests for those items. What are your thoughts on the bucket? It's the only item we haven't discussed yet.	
11	[A/GM] The bucket isn't very high on my priority list. If it's not very important to you either, perhaps we could simply decide who takes it based on our other allocations to keep things balanced. How does that sound?		
12		[GM/B] The bucket isn't very high on my priority list. If it's not very important to you either, perhaps we could simply decide who takes it based on our other allocations to keep things balanced. How does that sound?	
13			[GM/B] That sounds reasonable. Since you're interested in the chalk and nail, and I'm focusing on the pliers, we could add the bucket to one of our shares to balance things out. Would you like to propose how we might divide these items, including the bucket?
14		[A/GM] That sounds reasonable. Since you're interested in the chalk and nail, and I'm focusing on the pliers, we could add the bucket to one of our shares to balance things out. Would you like to propose how we might divide these items, including the bucket?	
15	[A/GM] Sure! Since I'm taking the chalk and nail, and you're interested in the pliers, how about I take the bucket as well to balance our shares? This way, you would get both pliers, and I would take the chalk, nail, and bucket. Does this distribution seem fair to you?		
16		[GM/B] Sure! Since I'm taking the chalk and nail, and you're interested in the pliers, how about I take the bucket as well to balance our shares? This way, you would get both pliers, and I would take the chalk, nail, and bucket. Does this distribution seem fair to you?	
17			[GM/B] Yes, that distribution sounds fair to me. You would take the chalk, nail, and bucket, and I would take the two pliers. Shall we proceed with this arrangement? If you agree, we can finalize this proposal.
18		[A/GM] Yes, that distribution sounds fair to me. You would take the chalk, nail, and bucket, and I would take the two pliers. Shall we proceed with this arrangement? If you agree, we can finalize this proposal.	
19	[A/GM] I agree with this arrangement. Let's finalize it. I'll make the secret proposal now.		
	[Proposal: 1 chalk, 1 nail, 1 bucket]		
20		[GM/B] The other player has submitted a secret proposal. It is now your turn to submit a proposal of your own.	
21			[GM/B] [Proposal: 2 pliers]

Figure 10: Example game episode generated through self-play using the GPT 4 model in the cooperative game settings. We can observe that even though the GPT 4 model was one of the best performing, especially for this game mode and language combination, this example still shows many deficiencies indicating a lack of strategy and grounding. First, already in the first message, player B misrepresents their value function, saying that they are particularly interested in the pliers, when they don't value them at all. It should be noted that in the cooperative setting it does not make any sense to lie to the other player. Furthermore, we can see in the dialogue that the players try to "balance things out", something that also does not make any sense in cooperative play.